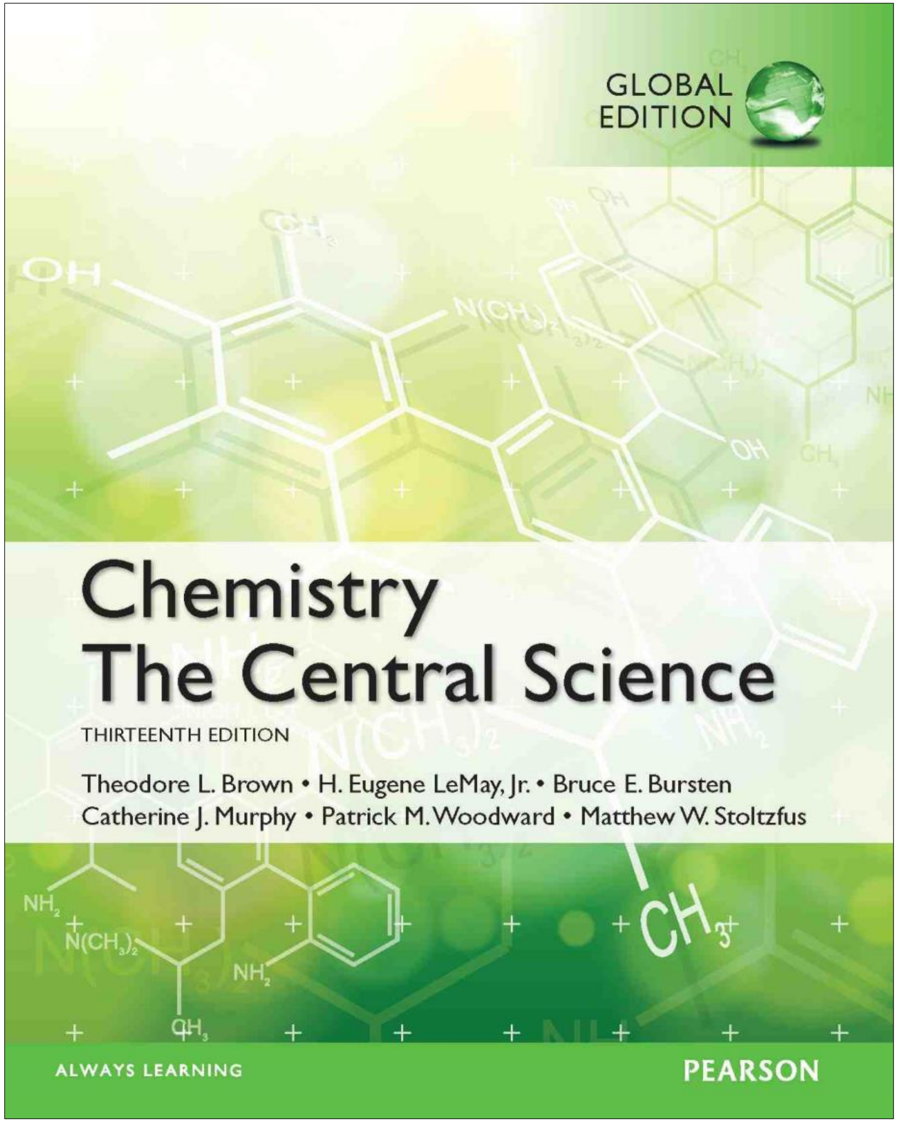

Lecture Presentation

Chapter 14

Chemical Kinetics

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Chemical Kinetics

Chemical Kinetics

- In chemical kinetics we study the rate (or speed) at which a chemical process occurs.

- Besides information about the speed at which reactions occur, kinetics also sheds light on the **reaction mechanism**, a molecular-level view of the path from reactants to products.

10^{-15} s

1 s

10^9 s
(30 years)

10^{15} s
(30 million years)

Time scale

Factors that Affect Reaction Rates

1) Physical state of the reactants
2) Reactant concentrations
3) Reaction temperature
4) Presence of a catalyst

Physical State of the Reactants

- The more readily the reactants collide, the more rapidly they react.
- Homogeneous reactions are often faster.
- Heterogeneous reactions that involve solids are faster if the surface area is increased; i.e., a fine powder reacts faster than a pellet or tablet.

Reactant Concentrations

- Increasing reactant concentration generally increases reaction rate.

- Since there are more molecules, more collisions occur.

Steel wool heated in air (about 20% O_2) glows red-hot but oxidizes to Fe_2O_3 slowly

Red-hot steel wool in 100% O_2 burns vigorously, forming Fe_2O_3 quickly

Temperature

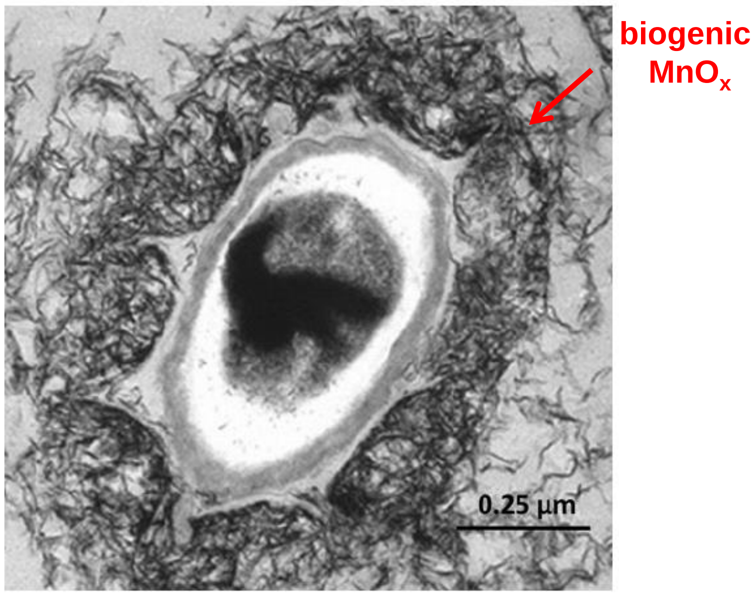
- Reaction rate generally increases with increased temperature.
- Kinetic energy of molecules is related to temperature.
- At higher temperatures, molecules move more quickly, increasing numbers of collisions and the energy the molecules possess during the collisions.

Presence of a Catalyst

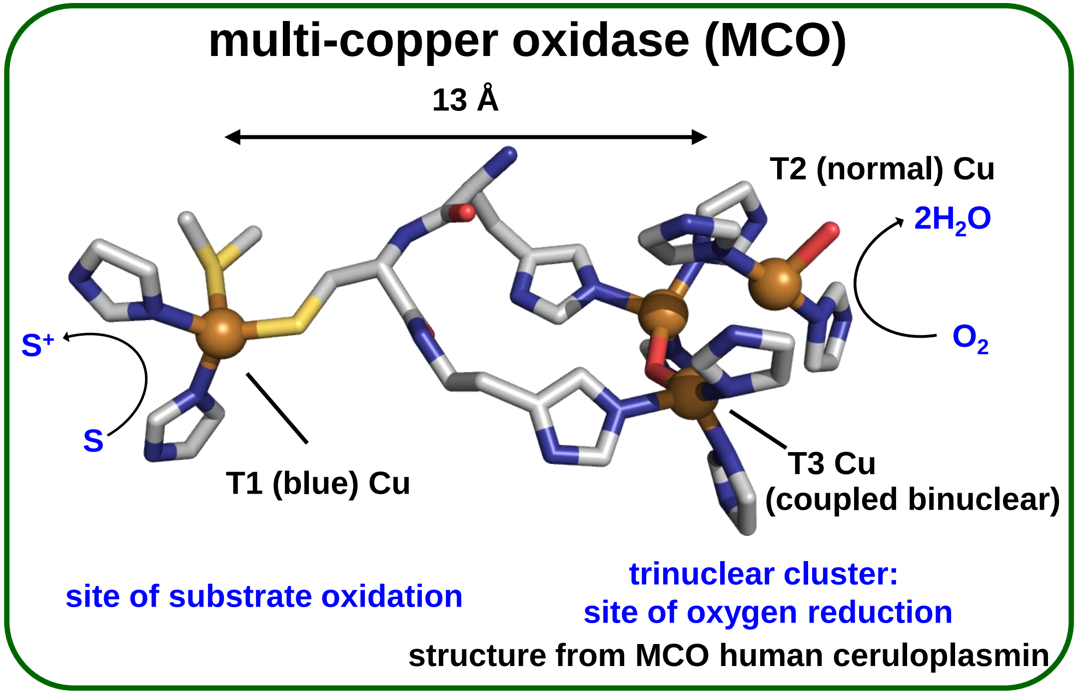
- Catalysts affect rate without being in the overall balanced equation.
- Catalysts affect the kinds of collisions, changing the mechanism (individual reactions that are part of the pathway from reactants to products).
- Catalysts are critical in many biological reactions.

Ocean bacterial Mn(II) oxidase MnxG

$Mn^{2+}(aq) \xrightarrow{?} MnO_x$



isolated from marine sediment off Scripps pier, 1980



Mnx protein complex

(mnxE_3F_3G, ~230 kDa)

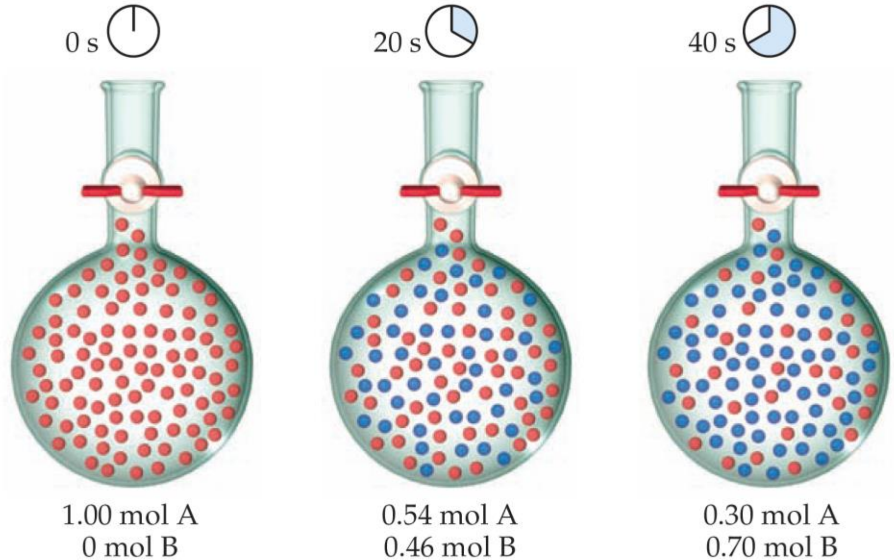
MnxG is the active MCO subunit

Chemical Kinetics

Tebo, et al., *Proc. Natl. Acad. Sci.*, 2013, 110, 11731.

Reaction Rate

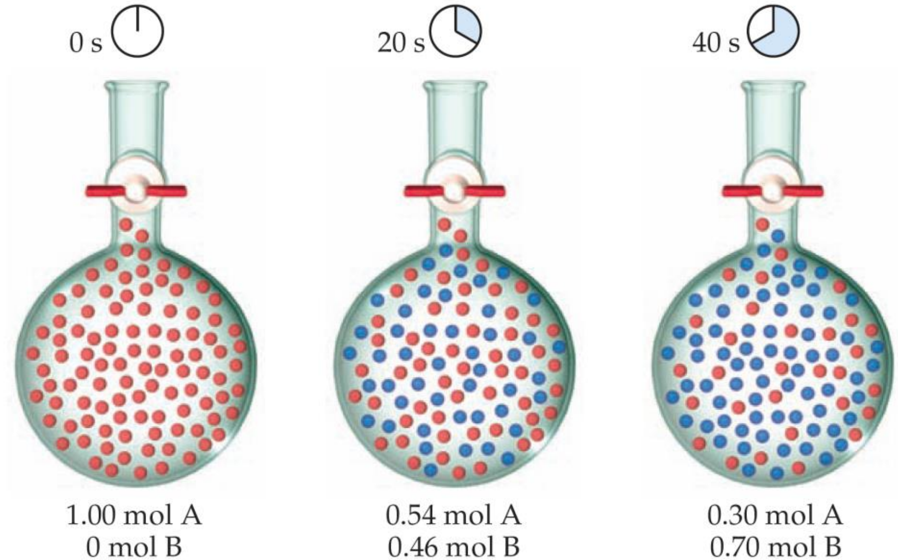
- **Rate** is a change in concentration over a time period: Δ[ ]/Δ*t*.
- Δ means “change in.”
- [ ] means molar concentration.
- *t* represents time.
- Types of rate measured:
  - average rate
  - instantaneous rate
  - initial rate



▲ **Figure 14.3** **Progress of a hypothetical reaction A $\longrightarrow$ B.** The volume of the flask is 1.0 L.

$$\text{Average rate of appearance of B} = \frac{\text{change in concentration of B}}{\text{change in time}}$$

$$= \frac{[\mathrm{B}] \text{ at } t_2 - [\mathrm{B}] \text{ at } t_1}{t_2 - t_1} = \frac{\Delta[\mathrm{B}]}{\Delta t}$$



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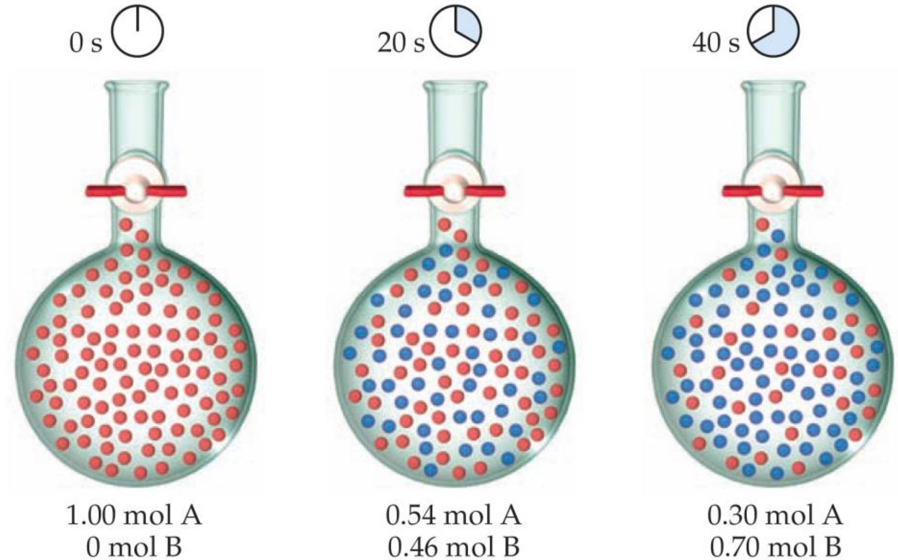
$$\text{Average rate} = \frac{0.46\,M - 0.00\,M}{20\text{ s} - 0\text{ s}} = 2.3 \times 10^{-2}\,M/\text{s}$$



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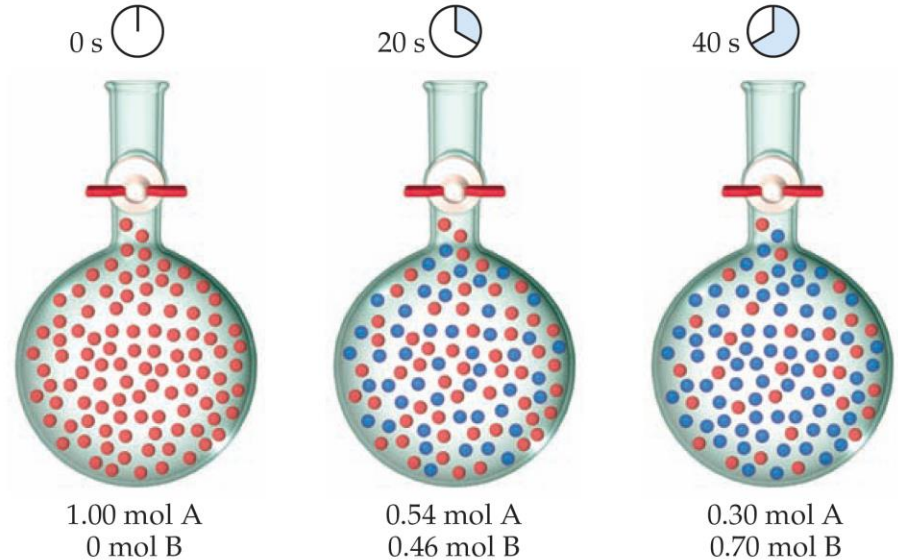
$$\text{Average rate of disappearance of A} = -\frac{\text{change in concentration of A}}{\text{change in time}}$$

$$= -\frac{\Delta[\text{A}]}{\Delta t}$$

Chemical Kinetics



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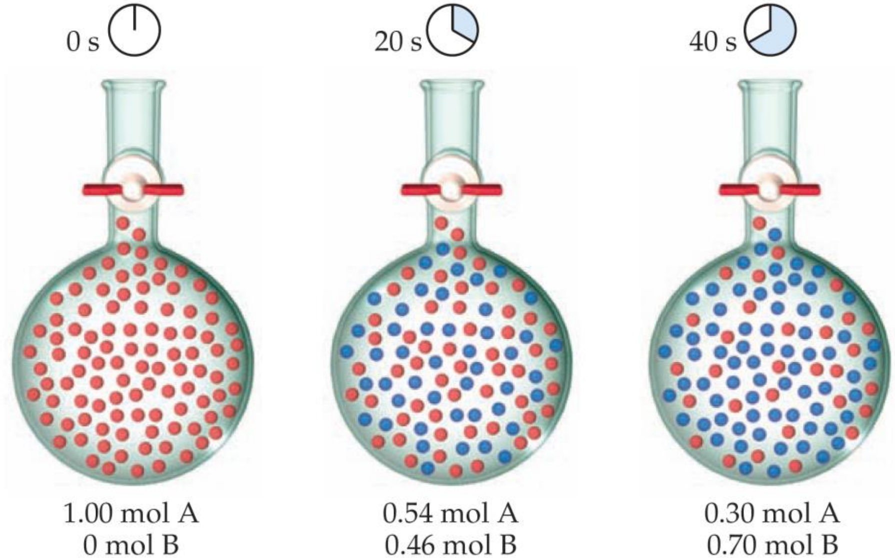
$$\text{Average rate of disappearance of A} = -\frac{\text{change in concentration of A}}{\text{change in time}}$$

$$= -\frac{\Delta[\text{A}]}{\Delta t}$$

$$\text{Average rate} = -\frac{\Delta[\text{A}]}{\Delta t} = -\frac{0.54\,M - 1.00\,M}{20\text{ s} - 0\text{ s}} = 2.3 \times 10^{-2}\,M/\text{s}$$

SAMPLE EXERCISE 14.1 Calculating an Average Rate of Reaction

From the data in Figure 14.3, calculate the average rate at which A disappears over the time interval from 20 s to 40 s.



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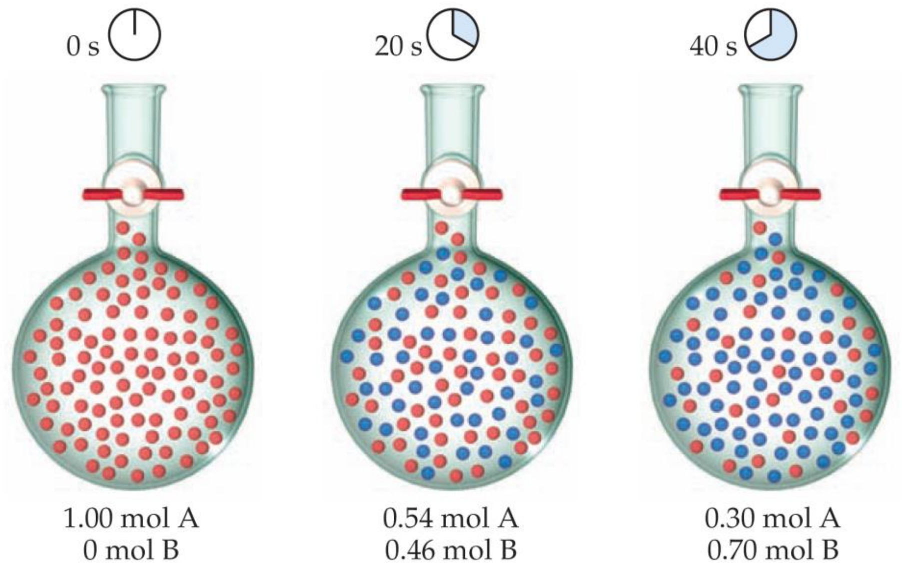

▲ **Figure 14.3 Progress of a hypothetical reaction A $\longrightarrow$ B.** The volume of the flask is 1.0 L.

Following Reaction Rates

$$C_4H_9Cl(aq) + H_2O(l) \longrightarrow C_4H_9OH(aq) + HCl(aq)$$

Table 14.1 Rate Data for Reaction of C_4H_9Cl with Water

Time, t (s)	$[C_4H_9Cl]$ (M)	Average Rate (M/s)
0.0	0.1000	
		1.9×10^{-4}
50.0	0.0905	
		1.7×10^{-4}
100.0	0.0820	
		1.6×10^{-4}
150.0	0.0741	
		1.4×10^{-4}
200.0	0.0671	
		1.22×10^{-4}
300.0	0.0549	
		1.01×10^{-4}
400.0	0.0448	
		0.80×10^{-4}
500.0	0.0368	
		0.560×10^{-4}
800.0	0.0200	
10,000	0	

- Rate of a reaction is measured using the concentration of a reactant or a product over time.
- In this example, $[C_4H_9Cl]$ is followed.

Following Reaction Rates

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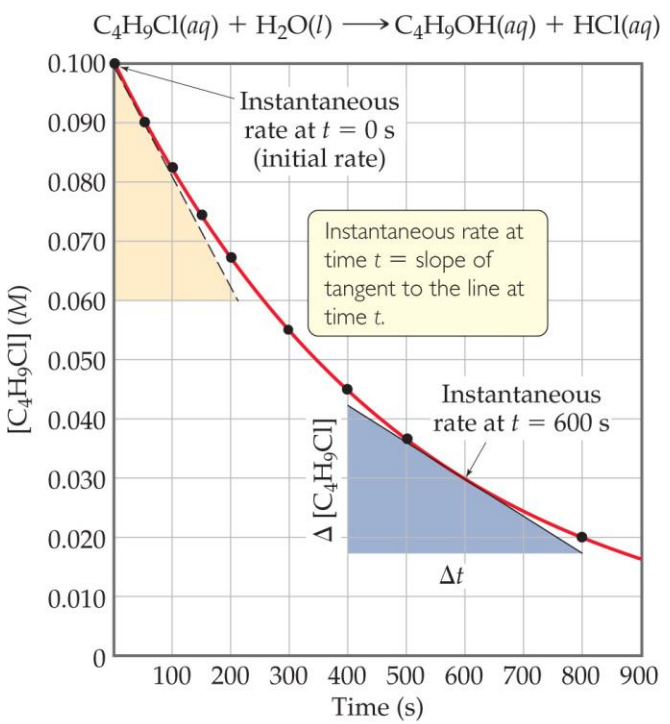
500.0	0.0368	
		0.560×10^{-4}
800.0	0.0200	
10,000	0	

- The average rate is calculated by the –(change in $[C_4H_9Cl]$) ÷ (change in time).
- The table shows the average rate for a variety of time intervals.

Plotting Rate Data

- A plot of the data gives more information about rate.
- The slope of the curve at one point in time gives the **instantaneous rate**.
- The instantaneous rate at time zero is called the **initial rate**; this is often the rate of interest to chemists.
- This figure shows instantaneous and initial rate of the earlier example.



Note: Reactions typically slow down over time!

Plotting Rate Data

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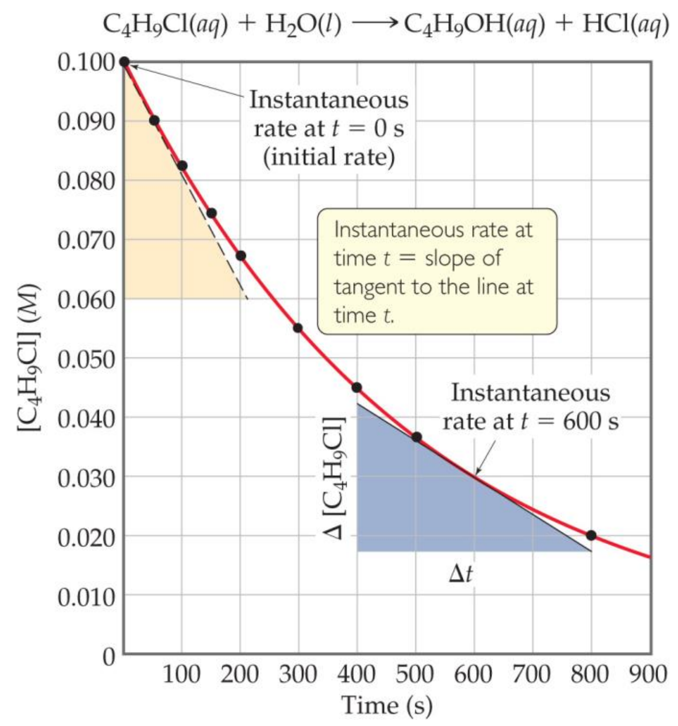

$$\text{Instantaneous rate} = -\frac{\Delta[C_4H_9Cl]}{\Delta t} = -\frac{(0.017 - 0.042)\,M}{(800 - 400)\,s}$$

$$= 6.3 \times 10^{-5}\,M/s$$

Plotting Rate Data

- A plot of the data gives more information about rate.
- The slope of the curve at one point in time gives the **instantaneous rate**.
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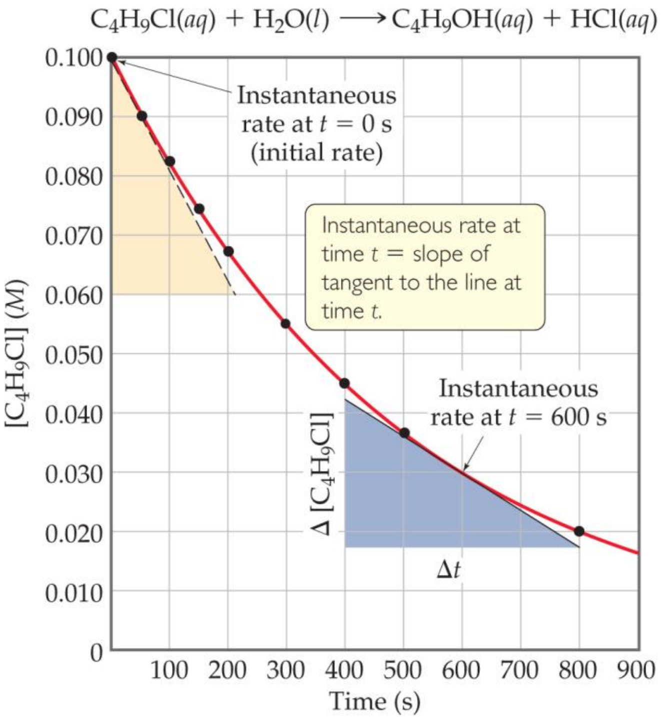


$$\text{Rate} = -\frac{\Delta[C_4H_9Cl]}{\Delta t} = -\frac{(0.060 - 0.100)\,M}{(210 - 0)\,\text{s}} = 1.9 \times 10^{-4}\,M/\text{s}$$

Chemical Kinetics

Relative Rates

- As was said, rates are followed using a reactant *or* a product. Does this give the same rate for each reactant and product?
- Rate is dependent on stoichiometry.
- If we followed *use* of C_4H_9Cl and compared it to *production* of C_4H_9OH, the values would be the same. Note that the *change* would have opposite signs—one goes down in value, the other goes up.

$$\text{Rate} = -\frac{\Delta[C_4H_9Cl]}{\Delta t} = \frac{\Delta[C_4H_9OH]}{\Delta t}$$

Relative Rates and Stoichiometry

- What if the equation is *not* 1:1?

- What will the relative rates be for:

  $2\ O_3\ (g) \rightarrow 3\ O_2\ (g)$

$$\text{Rate} = -\frac{1}{2}\frac{\Delta[O_3]}{\Delta t} = \frac{1}{3}\frac{\Delta[O_2]}{\Delta t}$$

Relative Rates and Stoichiometry

$$a\,\mathrm{A} + b\,\mathrm{B} \longrightarrow c\,\mathrm{C} + d\,\mathrm{D}$$

$$\text{Rate} = -\frac{1}{a}\frac{\Delta[\mathrm{A}]}{\Delta t} = -\frac{1}{b}\frac{\Delta[\mathrm{B}]}{\Delta t} = \frac{1}{c}\frac{\Delta[\mathrm{C}]}{\Delta t} = \frac{1}{d}\frac{\Delta[\mathrm{D}]}{\Delta t}$$

SAMPLE EXERCISE 14.3 Relating Rates at Which Products Appear and Reactants Disappear

(a) How is the rate at which ozone disappears related to the rate at which oxygen appears in the reaction $2\ O_3(g) \longrightarrow 3\ O_2(g)$?

(b) If the rate at which O_2 appears, $\Delta[O_2]/\Delta t$, is $6.0 \times 10^{-5}\ M/s$ at a particular instant, at what rate is O_3 disappearing at this same time, $-\Delta[O_3]/\Delta t$?

Determining Concentration Effect on Rate

- How do we determine what effect the concentration of each reactant has on the rate of the reaction?

- We keep every concentration constant *except* for one reactant and see what happens to the rate. *Then*, we change a different reactant. We do this until we have seen how *each* reactant has affected the rate.

An Example of How Concentration Affects Rate

$$NH_4^+(aq) + NO_2^-(aq) \longrightarrow N_2(g) + 2\,H_2O(l)$$

- Experiments 1–3 show how $[NH_4^+]$ affects rate.
- Experiments 4–6 show how $[NO_2^-]$ affects rate.
- Result: The **rate law**, which shows the relationship between rate and concentration for all reactants:

Rate = k $[NH_4^+]^a$ $[NO_2^-]^b$ **(rate law)**

Table 14.2 Rate Data for the Reaction of Ammonium and Nitrite Ions in Water at 25 °C

Experiment Number	Initial NH_4^+ Concentration (M)	Initial NO_2^- Concentration (M)	Observed Initial Rate (M/s)
1	0.0100	0.200	5.4×10^{-7}
2	0.0200	0.200	10.8×10^{-7}
3	0.0400	0.200	21.5×10^{-7}
4	0.200	0.0202	10.8×10^{-7}
5	0.200	0.0404	21.6×10^{-7}
6	0.200	0.0808	43.3×10^{-7}

More about Rate Law

- The exponents tell the **order** of the reaction with respect to each reactant.
- In our example from the last slide:

  Rate = k $[NH_4^+]$ $[NO_2^-]$
- The order with respect to each reactant is 1. (It is *first order* in NH_4^+ and NO_2^-.)
- The *reaction* is *second order* (1 + 1 = 2; we just add up all of the reactants' orders to get the reaction's order).
- What is k? It is the **rate constant**. It is a temperature-dependent quantity.

Rate Law: reaction order

$$a\,\mathrm{A} + b\,\mathrm{B} \longrightarrow c\,\mathrm{C} + d\,\mathrm{D}$$

$$\text{Rate} = k[\mathrm{A}]^m[\mathrm{B}]^n$$

$$2\,\mathrm{N_2O_5}(g) \longrightarrow 4\,\mathrm{NO_2}(g) + \mathrm{O_2}(g) \quad \text{Rate} = k[\mathrm{N_2O_5}] \qquad [14.9]$$

$$\mathrm{H_2}(g) + \mathrm{I_2}(g) \longrightarrow 2\,\mathrm{HI}(g) \quad \text{Rate} = k[\mathrm{H_2}][\mathrm{I_2}] \qquad [14.10]$$

$$\mathrm{CHCl_3}(g) + \mathrm{Cl_2}(g) \longrightarrow \mathrm{CCl_4}(g) + \mathrm{HCl}(g) \quad \text{Rate} = k[\mathrm{CHCl_3}][\mathrm{Cl_2}]^{1/2} \qquad [14.11]$$

SAMPLE EXERCISE 14.6 Determining a Rate Law from Initial Rate Data

The initial rate of a reaction $A + B \longrightarrow C$ was measured for several different starting concentrations of A and B, and the results are as follows:

Experiment Number	[A] (M)	[B] (M)	Initial Rate (M/s)
1	0.100	0.100	4.0×10^{-5}
2	0.100	0.200	4.0×10^{-5}
3	0.200	0.100	16.0×10^{-5}

Using these data, determine (**a**) the rate law for the reaction, (**b**) the rate constant, (**c**) the rate of the reaction when $[A] = 0.050 M$ and $[B] = 0.100 M$.

First Order Reactions

- Some rates depend *only* on one reactant to the first power.
- These are *first order* reactions.
- The rate law becomes:

$$\text{Rate} = k\,[\text{A}]$$

Relating *k* to [A] in a First Order Reaction

- rate = $k\,[A]$ (rate law)
- rate = $-\Delta\,[A]\,/\,\Delta t$ (rate expression)
- So: $k\,[A] = -\Delta\,[A]\,/\,\Delta t$
- Rearrange to: $\Delta\,[A]\,/\,[A] = -\,k\,\Delta t$
- Integrate: $\ln\,([A]\,/\,[A]_o) = -\,k\,t$
- Rearrange: $\ln\,[A] = -\,k\,\mathrm{t} + \ln\,[A]_o$
- Note: this follows the equation of a line:

$$y = m\,x + b$$

- So, a plot of $\ln\,[A]$ vs. t is linear.

Chemical Kinetics

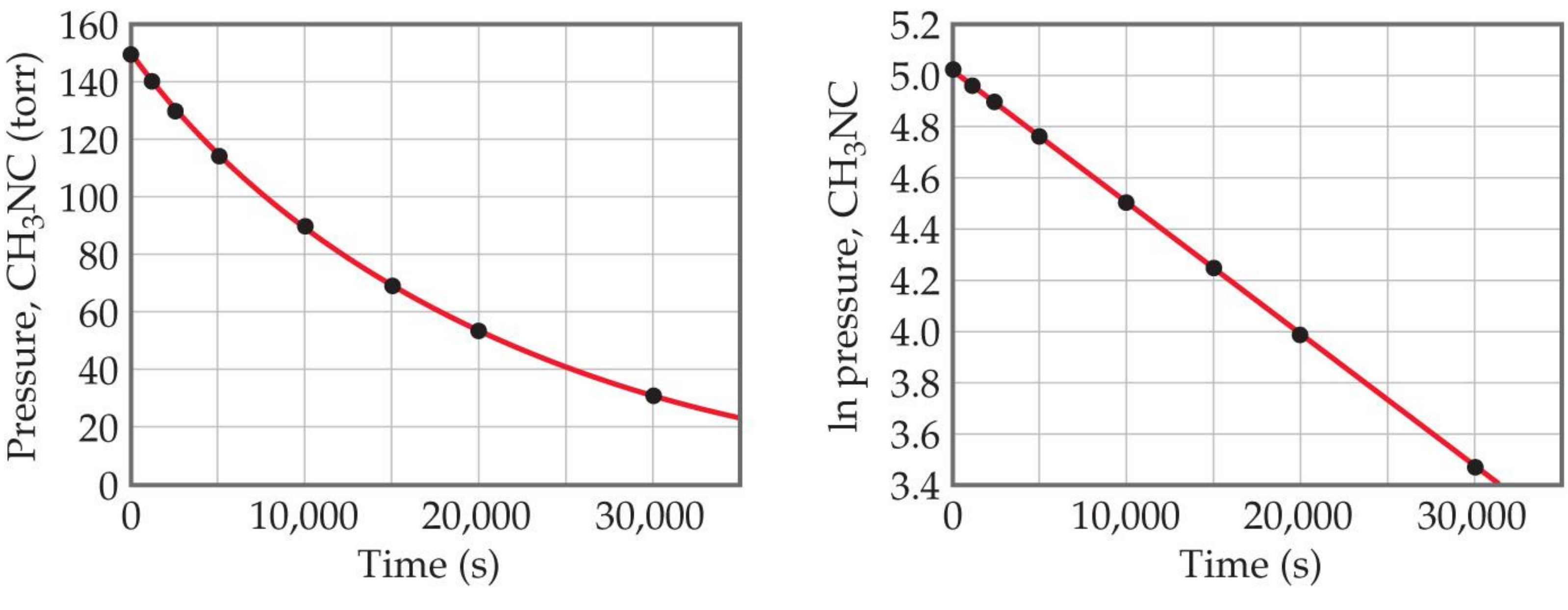
Ex: Conversion of Methyl Isonitrile to Acetonitrile

- The equation for the reaction:

$$CH_3NC \rightarrow CH_3CN$$

- It is first order.

$$\text{Rate} = k\,[CH_3NC]$$



(a) (b)

Finding the Rate Constant, *k*

- Besides using the rate law, we can find the rate constant from the plot of ln [*A*] vs. *t*.

- Remember the integrated rate law:

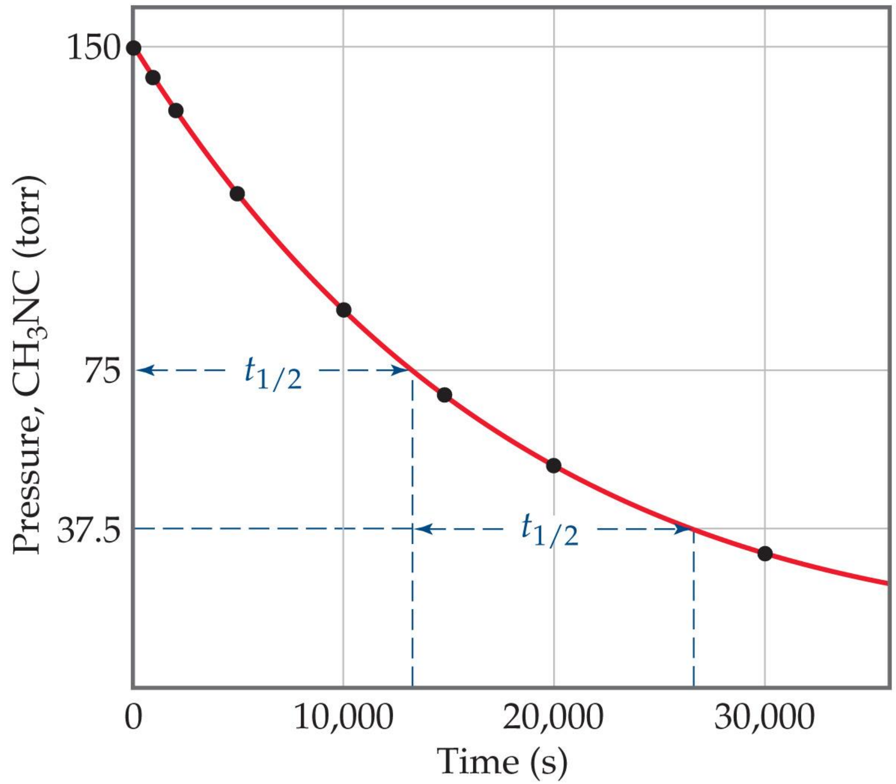
$$\ln [A] = -k\,t + \ln [A]_o$$

- The plot will give a line. Its slope will equal −*k*.

Half-life ($t_{1/2}$)

- Definition: The amount of time it takes for one-half of a reactant to be used up in a chemical reaction.

- First Order Reaction:
  - $\ln [A] = -k\,t + \ln [A]_o$
  - $\ln ([A]_o/2) = -k\,t_{1/2} + \ln [A]_o$
  - $-\ln ([A]_o/2) + \ln [A]_o = k\,t_{1/2}$
  - $\ln ([A]_o / [A]_o/2) = k\,t_{1/2}$
  - $\ln 2 = k\,t_{1/2}$ or $t_{1/2} = 0.693/k$



Chemical Kinetics

SAMPLE EXERCISE 14.7 Using the Integrated First-Order Rate Law

The decomposition of a certain insecticide in water at 12 °C follows first-order kinetics with a rate constant of $1.45\ yr^{-1}$. A quantity of this insecticide is washed into a lake on June 1, leading to a concentration of $5.0 \times 10^{-7}\ g/cm^3$. Assume that the temperature of the lake is constant (so that there are no effects of temperature variation on the rate). (**a**) What is the concentration of the insecticide on June 1 of the following year? (**b**) How long will it take for the insecticide concentration to decrease to $3.0 \times 10^{-7}\ g/cm^3$?

Second Order Reactions

- Some rates depend *only* on a reactant to the second power.

- These are *second order* reactions.

- The rate law becomes:

$$\text{Rate} = k\,[A]^2$$

Solving the Second Order Reaction for

A → Products

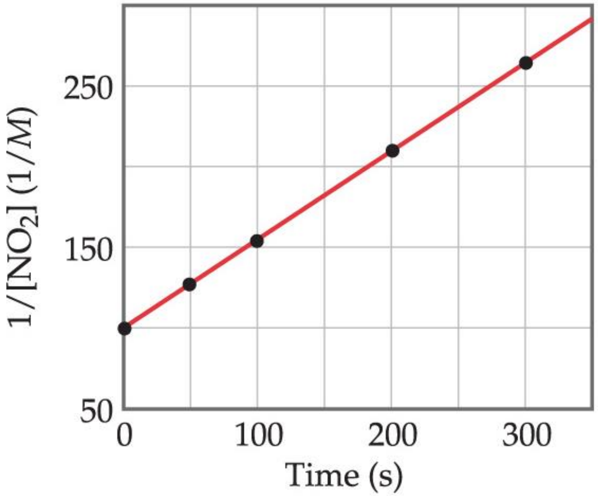
- rate = $k\,[A]^2$
- rate = $-\Delta\,[A]\,/\,\Delta\,t$
- So, $k\,[A]^2 = -\Delta\,[A]\,/\,\Delta\,t$
- Rearranging: $\Delta\,[A]\,/\,[A]^2 = -k\;\Delta\,t$
- Using calculus: $1/[A] = 1/[A]_o + k\,t$
- Notice: The linear relationships for first order and second order reactions differ!

Chemical Kinetics

An Example of a Second Order Reaction: Decomposition of NO_2

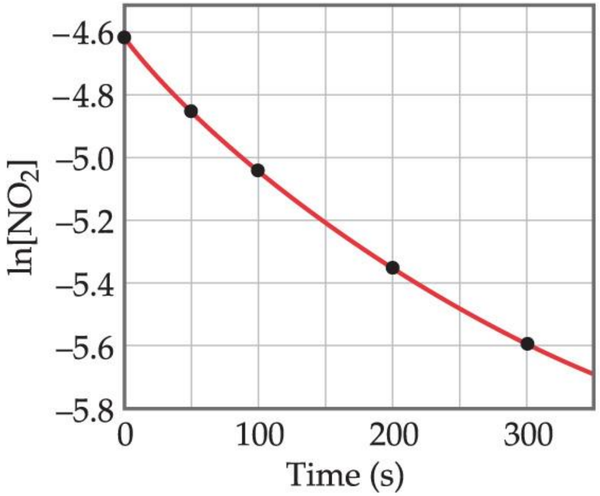
$1/[A] = 1/[A]_o + k\,t$

➢ A plot following NO_2 decomposition shows that it must be second order because it is linear for $1/[NO_2]$, *not* linear for $\ln [NO_2]$.



➢ Equation:

$NO_2 \rightarrow NO + ½\ O_2$



Chemical Kinetics

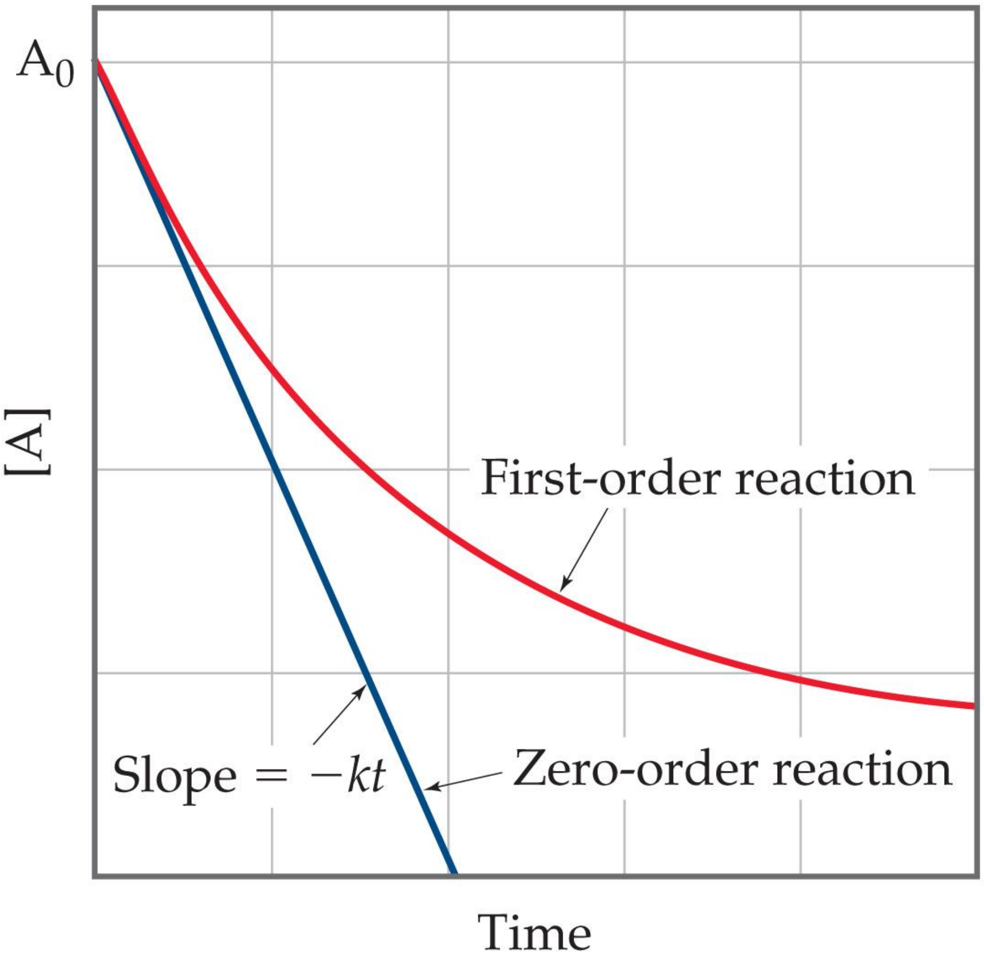
Half-Life and Second Order Reactions

- Using the integrated rate law, we can see how half-life is derived:
  - $1/[A] = 1/[A]_o + k\,t$
  - $1/([A]_o/2) = 1/[A]_o + k\,t_{½}$
  - $2/[A]_o - 1/[A]_o = k\,t_{½}$
  - $t_{½} = 1 / (k\,[A]_o)$
- So, half-life is a concentration-dependent quantity for second order reactions!

Zero Order Reactions

- Occasionally, rate is *independent* of the concentration of the reactant:
- Rate = k
- These are *zero order* reactions.

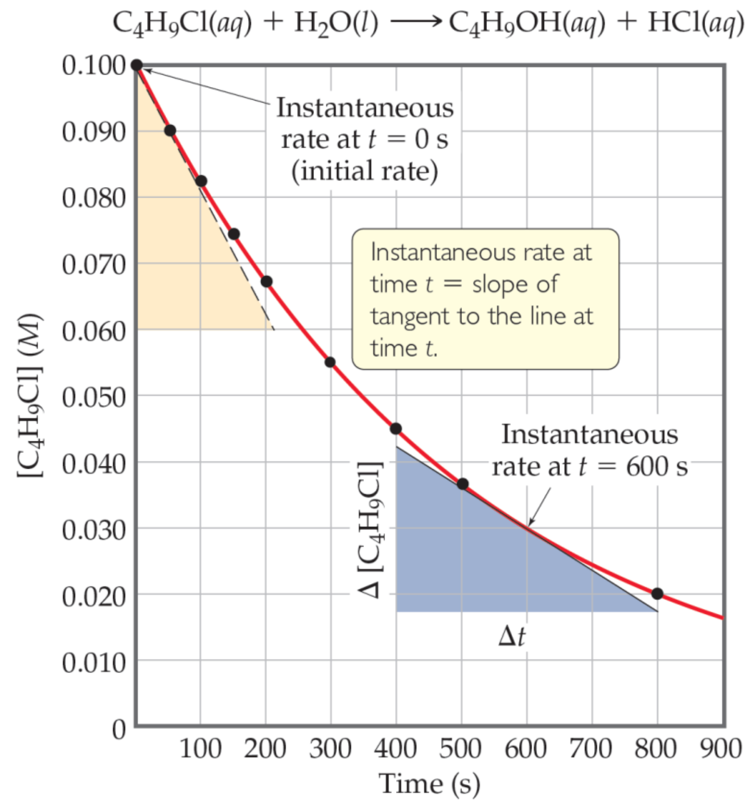
- These reactions are linear in concentration.



Chemical Kinetics

SAMPLE EXERCISE 14.9 Determining the Half-Life of a First-Order Reaction

The reaction of C_4H_9Cl with water is a first-order reaction. (**a**) Use Figure 14.4 to estimate the half-life for this reaction. (**b**) Use the half-life from (a) to calculate the rate constant.



▲ **Figure 14.4** **Concentration of butyl chloride (C_4H_9Cl) as a function of time.**

Chemical Kinetics

Factors That Affect Reaction Rate

1. Temperature

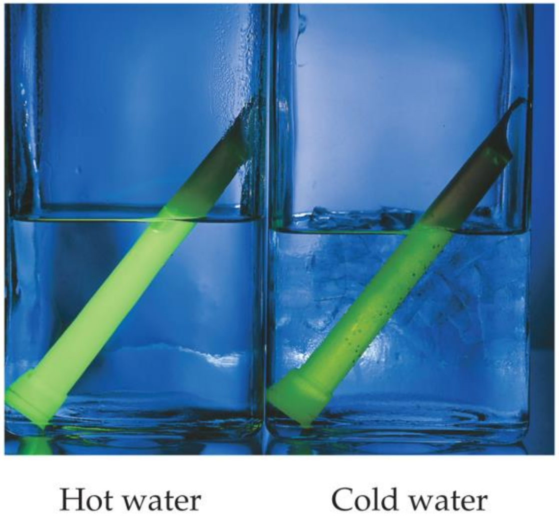
2. Frequency of collisions

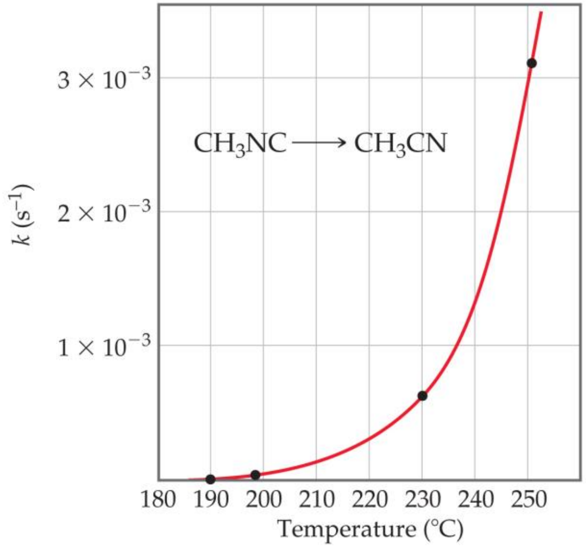
3. Orientation of molecules

4. Energy needed for the reaction to take place (activation energy)

Temperature and Rate

- Generally, as temperature increases, rate increases.
- The rate constant is temperature dependent: it increases as temperature increases.
- Rate constant doubles (approximately) with every 10 ºC rise.





Chemical Kinetics

The Collision Model

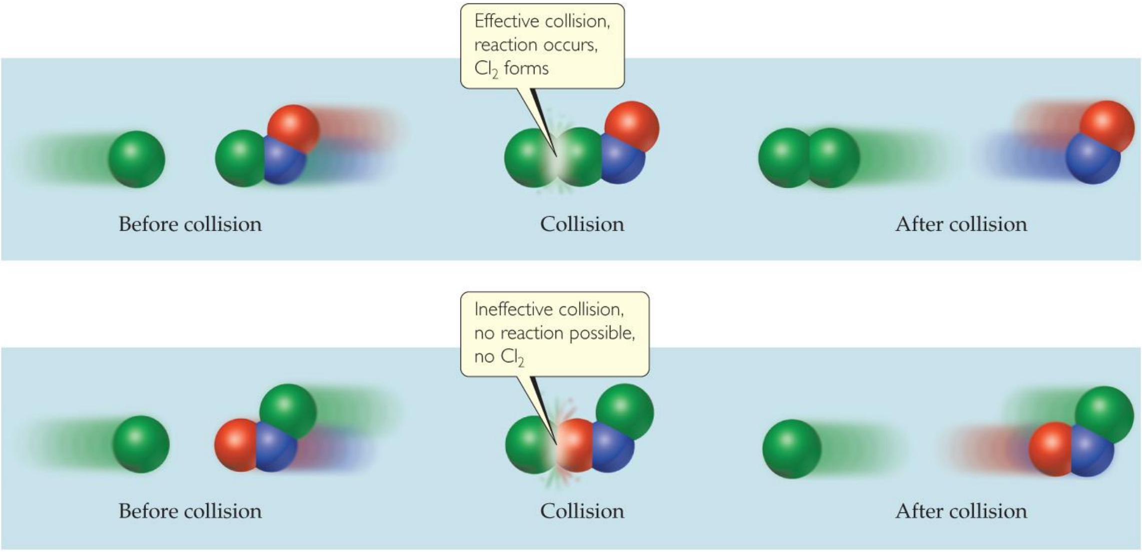
- In a chemical reaction, bonds are broken and new bonds are formed.
- Molecules can only react if they collide with each other.

Frequency of Collisions

- The **collision model** is based on the kinetic molecular theory.
- Molecules must collide to react.
- If there are more collisions, more reactions can occur.
- So, if there are more molecules, the reaction is faster.
- Also, if the temperature is higher, molecules move faster, causing more collisions and a higher rate of reaction.

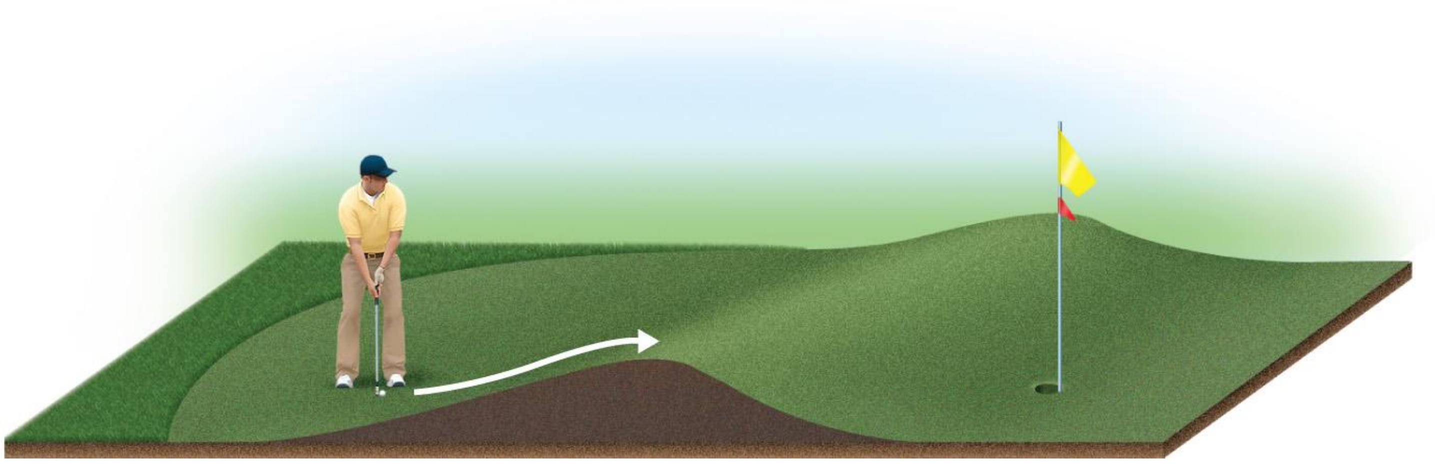
Orientation of Molecules

- Molecules can often collide without forming products.
- Aligning molecules properly can lead to chemical reactions.
- Bonds must be broken and made and atoms need to be in proper positions.



Energy Needed for a Reaction to Take Place (Activation Energy)

- The minimum energy needed for a reaction to take place is called **activation energy**.
- An energy barrier must be overcome for a reaction to take place, much like the ball must be hit to overcome the barrier in the figure below.



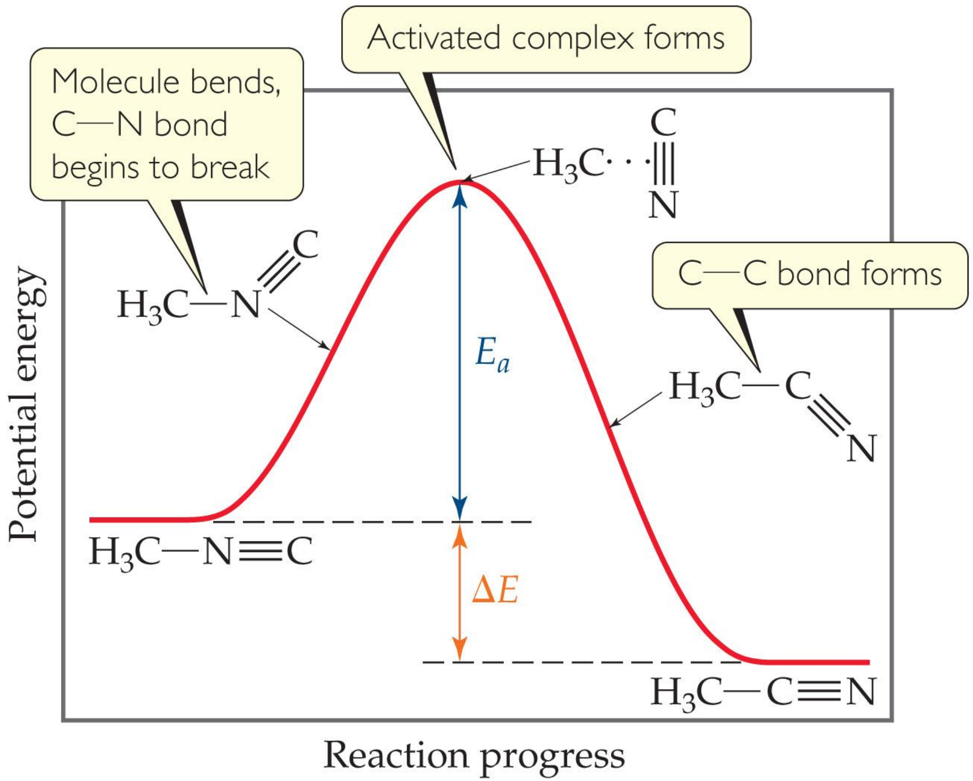
Chemical Kinetics

Transition State (Activated Complex)

- Reactants gain energy as the reaction proceeds until the particles reach the maximum energy state.
- The organization of the atoms at this highest energy state is called the **transition state** (or activated complex).
- The energy needed to form this state is called the *activation energy*.

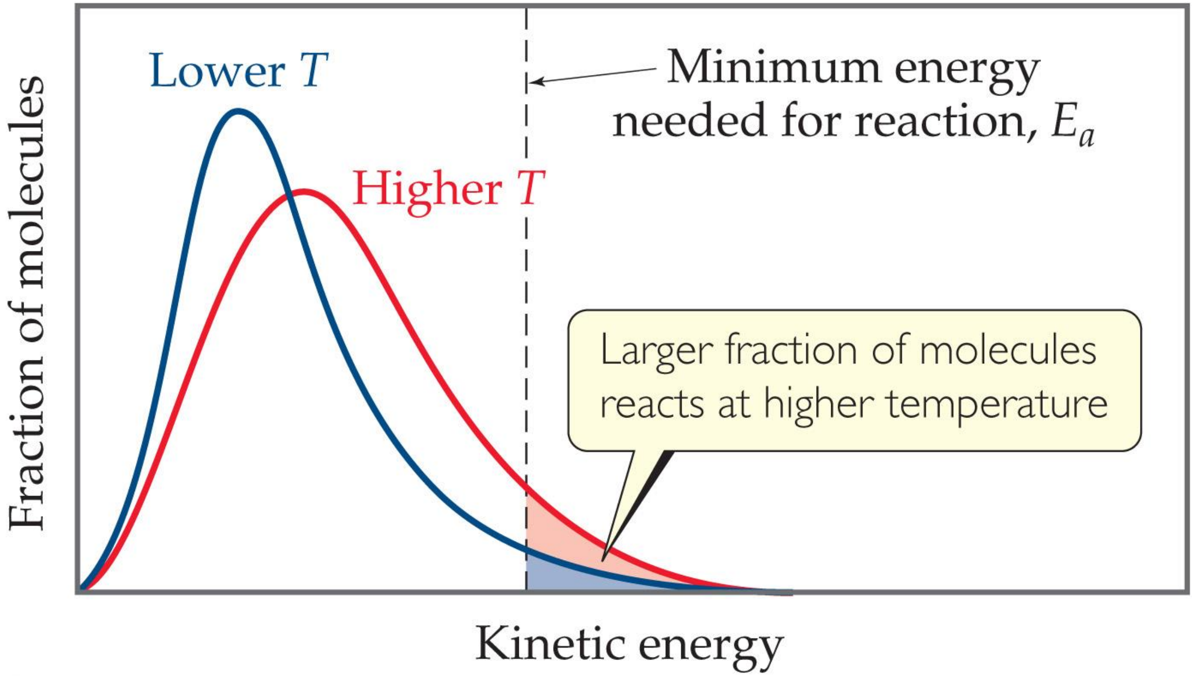
Reaction Progress

- Plots are made to show the energy possessed by the particles as the reaction proceeds.
- At the highest energy state, the transition state is formed.
- Reactions can be endothermic or exothermic after this.



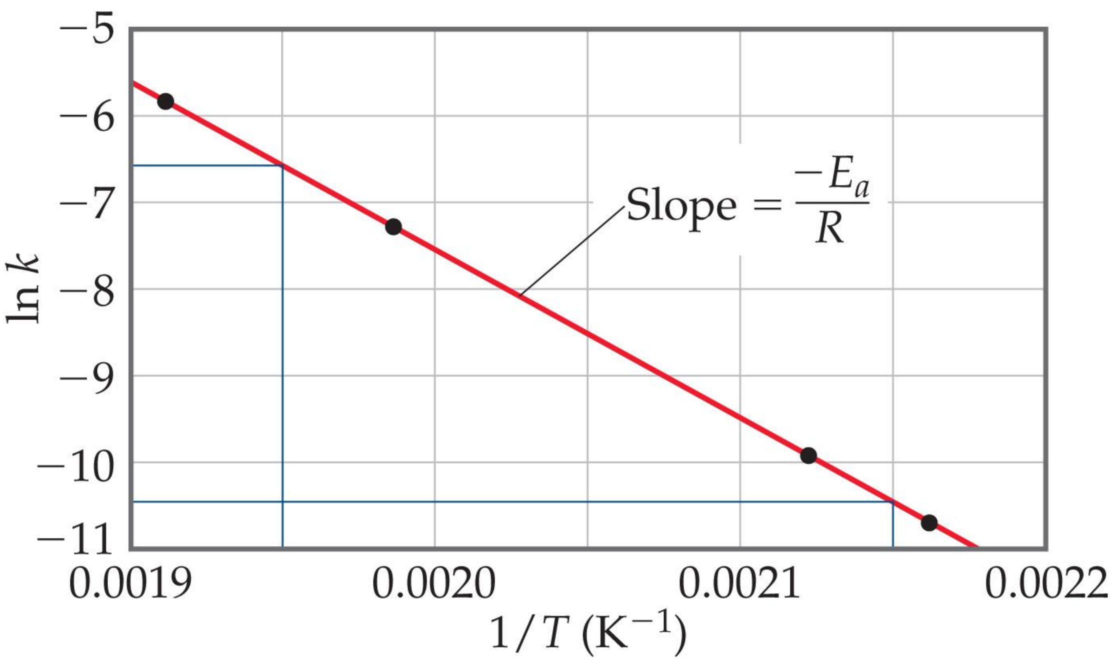
Distribution of the Energy of Molecules

- Gases have an average temperature, but each individual molecule has its own energy.
- At higher energies, more molecules possess the energy needed for the reaction to occur.



The Relationship Between Activation Energy & Temperature

- Arrhenius noted relationship between activation energy and temperature: $k = Ae^{-Ea/RT}$
- Activation energy can be determined graphically by reorganizing the equation: $\ln k = -E_a/RT + \ln A$



$$\ln \frac{k_1}{k_2} = \frac{E_a}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

Arrhenius Equation

- Arrhenius noted relationship between activation energy and temperature: $k = Ae^{-Ea/RT}$
- Activation energy can be determined graphically by reorganizing the equation: ln $k = -E_a/RT + \ln A$

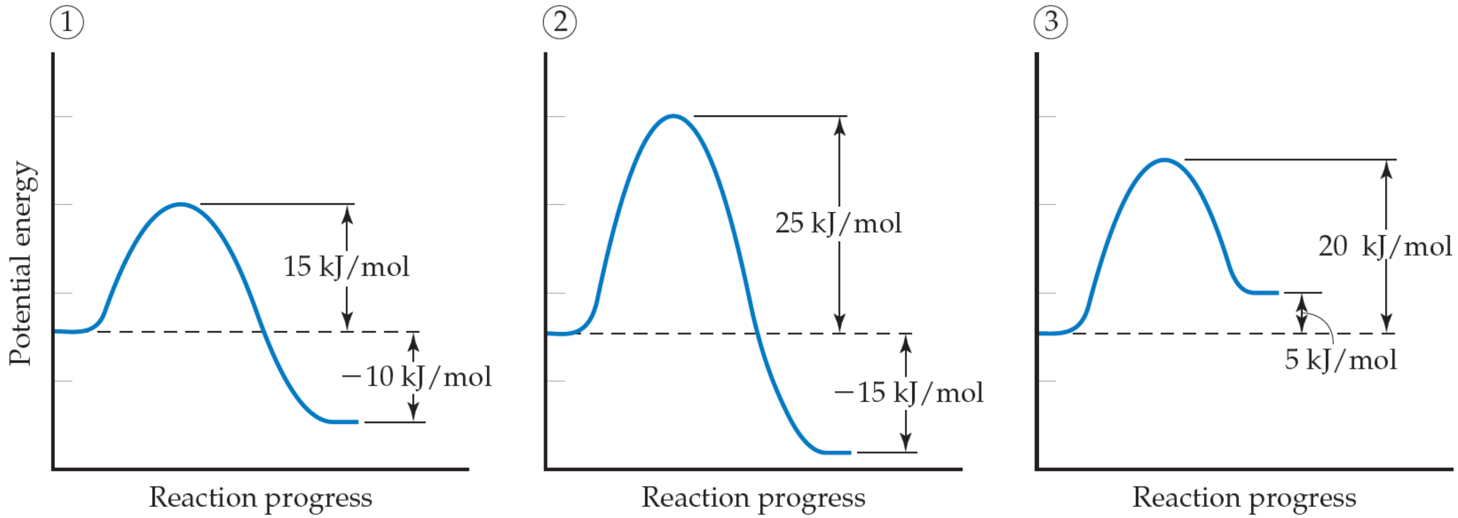
These three factors are incorporated into the **Arrhenius equation**:

$$k = Ae^{-E_a/RT} \qquad [14.19]$$

In this equation, k is the rate constant, E_a is the activation energy, R is the gas constant (8.314 J/mol-K), and T is the absolute temperature. The **frequency factor**, A, is constant, or nearly so, as temperature is varied. This factor is related to the frequency of collisions and the probability that the collisions are favorably oriented for reaction.* As the magnitude of E_a increases, k decreases because the fraction of molecules that possess the required energy is smaller. Thus, at fixed values of T and A, *reaction rates decrease as E_a increases.*

SAMPLE EXERCISE 14.10 Activation Energies and Speeds of Reaction

Consider a series of reactions having these energy profiles:



Chemical Kinetics

Law vs. Theory

- Kinetics gives *what* happens. We call the description the *rate law*.
- *Why (how)* do we observe that rate law? We explain with a *theory* called a **mechanism**.
- A mechanism is a series of stepwise reactions that show how reactants become products.

Reaction Mechanisms

- Reactions may occur all at once or through several discrete steps.
- Each of these processes is known as an **elementary reaction** or **elementary process**.

Molecularity

Table 14.3 Elementary Reactions and Their Rate Laws

Molecularity	Elementary Reaction	Rate Law
*Uni*molecular	$A \longrightarrow$ products	Rate $= k[A]$
*Bi*molecular	$A + A \longrightarrow$ products	Rate $= k[A]^2$
*Bi*molecular	$A + B \longrightarrow$ products	Rate $= k[A][B]$
*Ter*molecular	$A + A + A \longrightarrow$ products	Rate $= k[A]^3$
*Ter*molecular	$A + A + B \longrightarrow$ products	Rate $= k[A]^2[B]$
*Ter*molecular	$A + B + C \longrightarrow$ products	Rate $= k[A][B][C]$

The **molecularity** of an elementary reaction tells how many molecules are involved in that step of the mechanism.

Termolecular?

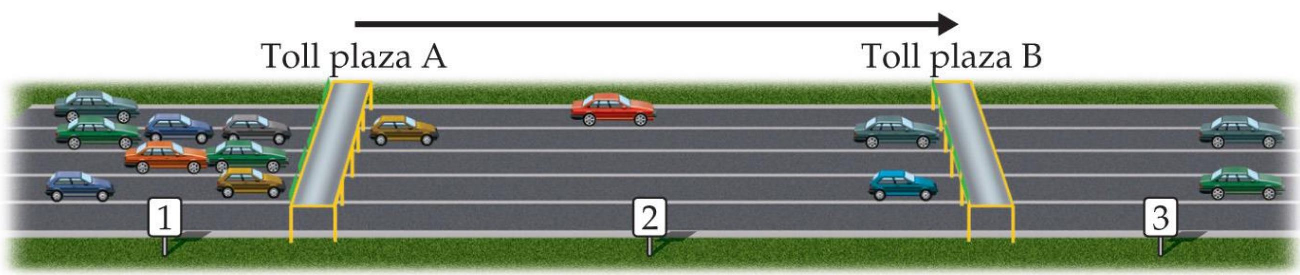
- Termolecular steps require three molecules to simultaneously collide with the proper orientation *and* the proper energy.

- These are rare, if they indeed do occur.
- These *must* be slower than unimolecular or bimolecular steps.

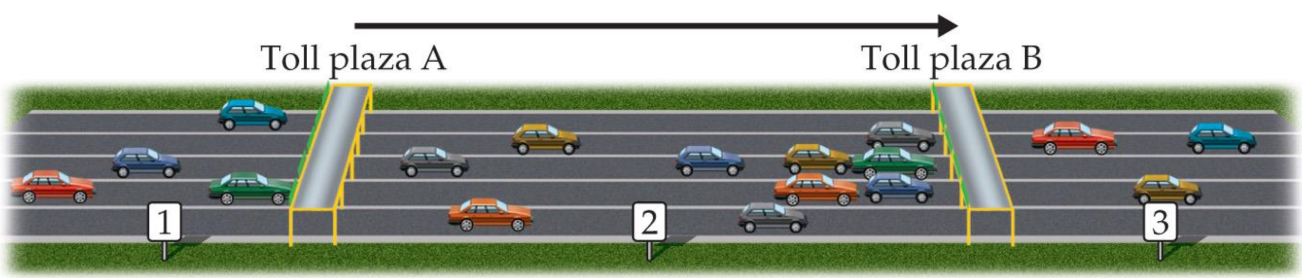
- Nearly *all* mechanisms use *only* unimolecular or bimolecular reactions.

What Limits the Rate?

- The overall reaction cannot occur faster than the slowest reaction in the mechanism.
- We call that the **rate-determining/limiting step**.



(a) Cars slowed at toll plaza A, rate-determining step is passage through A



(b) Cars slowed at toll plaza B, rate-determining step is passage through B

What is Required of a Plausible Mechanism?

- The *rate law* must be able to be derived from the rate-determining step.
- The *stoichiometry* must be obtained when all steps are added up.
- Each step must balance, like any equation.
- All intermediates are made and used up.
- Any catalyst is used and regenerated.

A Mechanism With a Slow Initial Step

- Overall equation: $NO_2 + CO \rightarrow NO + CO_2$
- Rate law: Rate = $k\,[NO_2]^2$

- If the first step is the rate-determining step, the coefficients on the reactants side are the same as the order in the rate law!

- So, the first step of the mechanism begins:

$$NO_2 + NO_2 \rightarrow$$

A Mechanism With a Slow Initial Step (continued)

- The easiest way to complete the first step is to make a product:

$$NO_2 + NO_2 \rightarrow NO + NO_3$$

- We do not see NO_3 in the stoichiometry, so it is an **intermediate**, which needs to be used in a faster next step.

$$NO_3 + CO \rightarrow NO_2 + CO_2$$

A Mechanism With a Slow Initial Step (completed)

- Since the first step is the slowest step, it gives the rate law.
- If you add up all of the individual steps (2 of them), you get the stoichiometry.
- Each step balances.
- This is a plausible mechanism.

SAMPLE EXERCISE 14.12 Determining Molecularity and Identifying Intermediates

It has been proposed that the conversion of ozone into O_2 proceeds by a two-step mechanism:

$$O_3(g) \longrightarrow O_2(g) + O(g)$$

$$O_3(g) + O(g) \longrightarrow 2\,O_2(g)$$

(a) Describe the molecularity of each elementary reaction in this mechanism.
(b) Write the equation for the overall reaction.
(c) Identify the intermediate(s).

SAMPLE EXERCISE 14.14 Determining the Rate Law for a Multistep Mechanism

The decomposition of nitrous oxide, N_2O, is believed to occur by a two-step mechanism:

$$N_2O(g) \longrightarrow N_2(g) + O(g) \quad \text{(slow)}$$

$$N_2O(g) + O(g) \longrightarrow N_2(g) + O_2(g) \quad \text{(fast)}$$

(**a**) Write the equation for the overall reaction. (**b**) Write the rate law for the overall reaction.

A Mechanism With a Fast Initial Step

- Equation for the reaction:

$$2\ NO + Br_2 \rightleftharpoons 2\ NOBr$$

- The rate law for this reaction is found to be

$$\text{Rate} = k\,[NO]^2\,[Br_2]$$

- Because termolecular processes are rare, this rate law suggests a multistep mechanism.

A Mechanism With a Fast Initial Step (continued)

- The rate law indicates that a quickly established equilibrium is followed by a slow step.

- Step 1: $NO + Br_2 \rightleftharpoons NOBr_2$
- Step 2: (rds) $NOBr_2 + NO \rightarrow 2\ NOBr$

What is the Rate Law?

- The rate of the overall reaction depends upon the rate of the slow step.

- The rate law for that step would be

$$\text{Rate} = k_2[\text{NOBr}_2]\ [\text{NO}]$$

- But how can we find $[\text{NOBr}_2]$?

[$NOBr_2$] (An Intermediate)?

- $NOBr_2$ can react two ways:
  - With NO to form NOBr.
  - By decomposition to reform NO and Br_2.
- The reactants and products of the first step are in equilibrium with each other.
- For an equilibrium (discussed in the next chapter):

$$\textbf{Rate}_{f} = \textbf{Rate}_{r}$$

The Rate Law (Finally!)

- Substituting for the forward and reverse rates:

$$k_1 \,[\mathrm{NO}]\,[\mathrm{Br_2}] = k_{-1}\,[\mathrm{NOBr_2}]$$

- Solve for $[\mathrm{NOBr_2}]$, then substitute into the rate law:

$$\text{Rate} = k_2\,(k_1/k_{-1})\,[\mathrm{NO}]\,[\mathrm{Br_2}]\,[\mathrm{NO}]$$

- This gives the observed rate law!

$$\text{Rate} = k\,[\mathrm{NO}]^2\,[\mathrm{Br_2}]$$

SAMPLE EXERCISE 14.15 Deriving the Rate Law for a Mechanism with a Fast Initial Step

Show that the following mechanism for Equation 14.24 also produces a rate law consistent with the experimentally observed one:

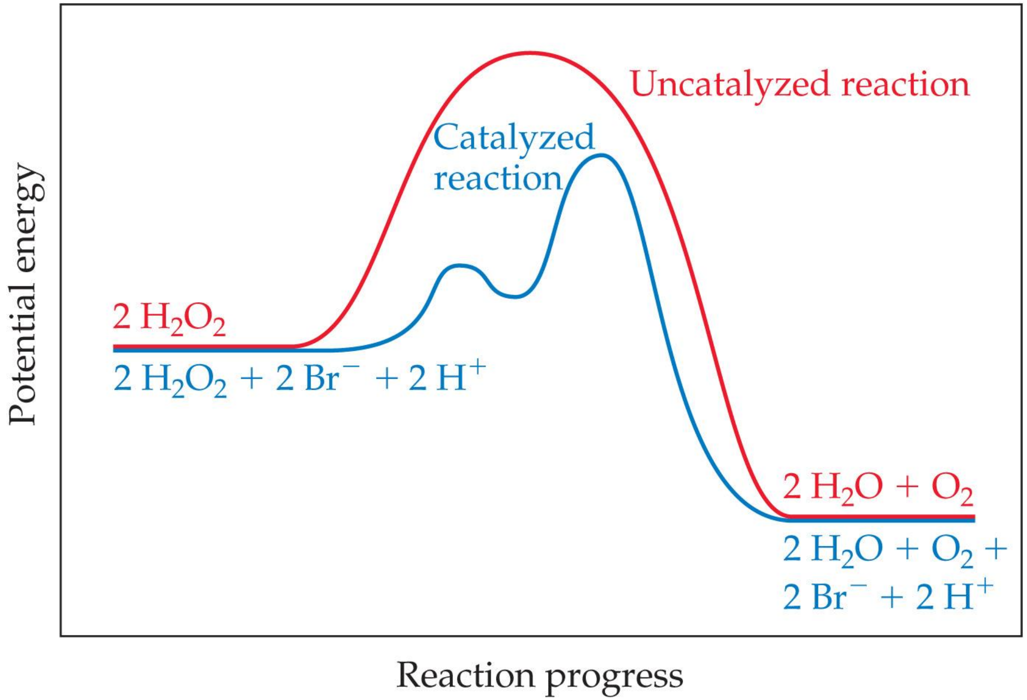
$$\text{Step 1:}\quad NO(g) + NO(g) \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} N_2O_2(g) \quad \text{(fast, equilibrium)}$$

$$\text{Step 2:}\quad N_2O_2(g) + Br_2(g) \xrightarrow{k_2} 2\,NOBr(g) \quad \text{(slow)}$$

Catalysts

- Catalysts increase the rate of a reaction by decreasing the activation energy of the reaction.

- Catalysts change the mechanism by which the process occurs.



Chemical Kinetics

Types of Catalysts

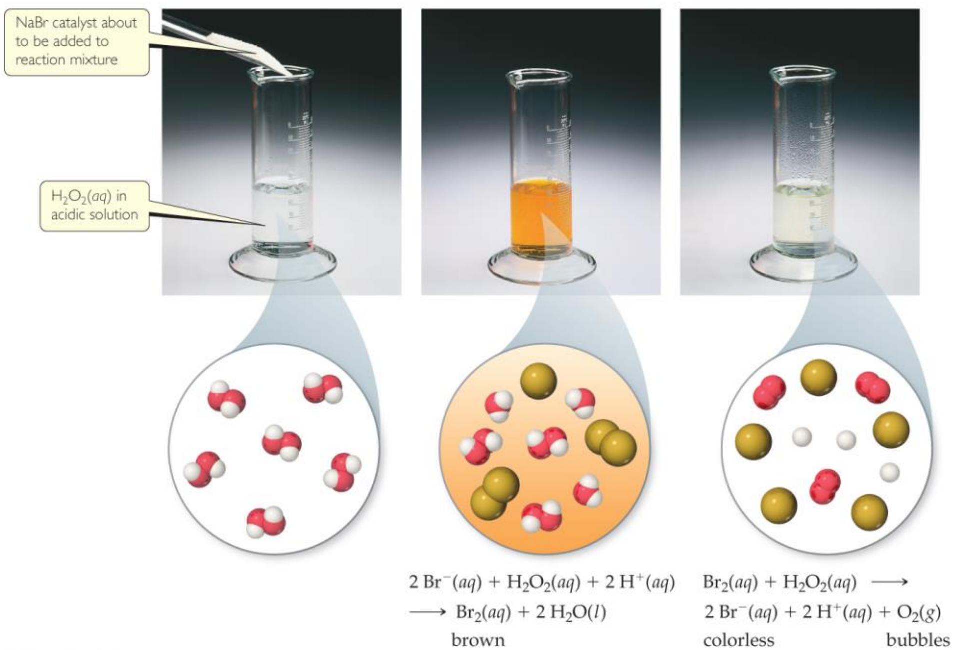
1) Homogeneous catalysts

2) Heterogeneous catalysts

3) Enzymes (catalysts in biological system)

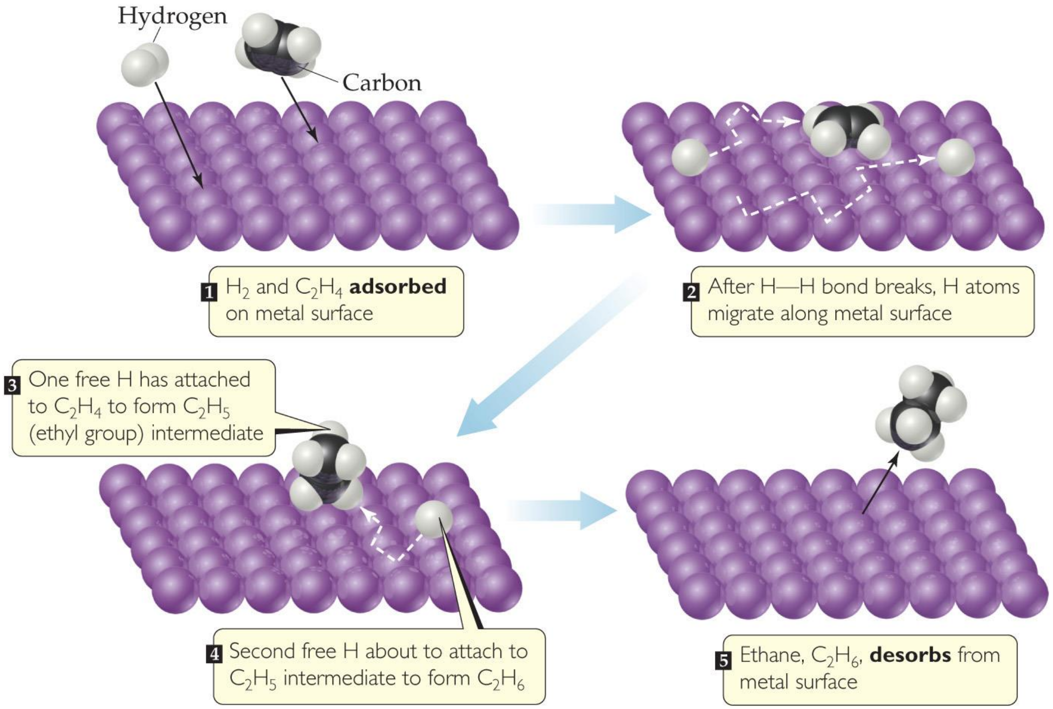
Homogeneous Catalysts

- The reactants and catalyst are in the same phase.
- Many times, reactants and catalyst are dissolved in the same solvent, as seen below.



Heterogeneous Catalysts

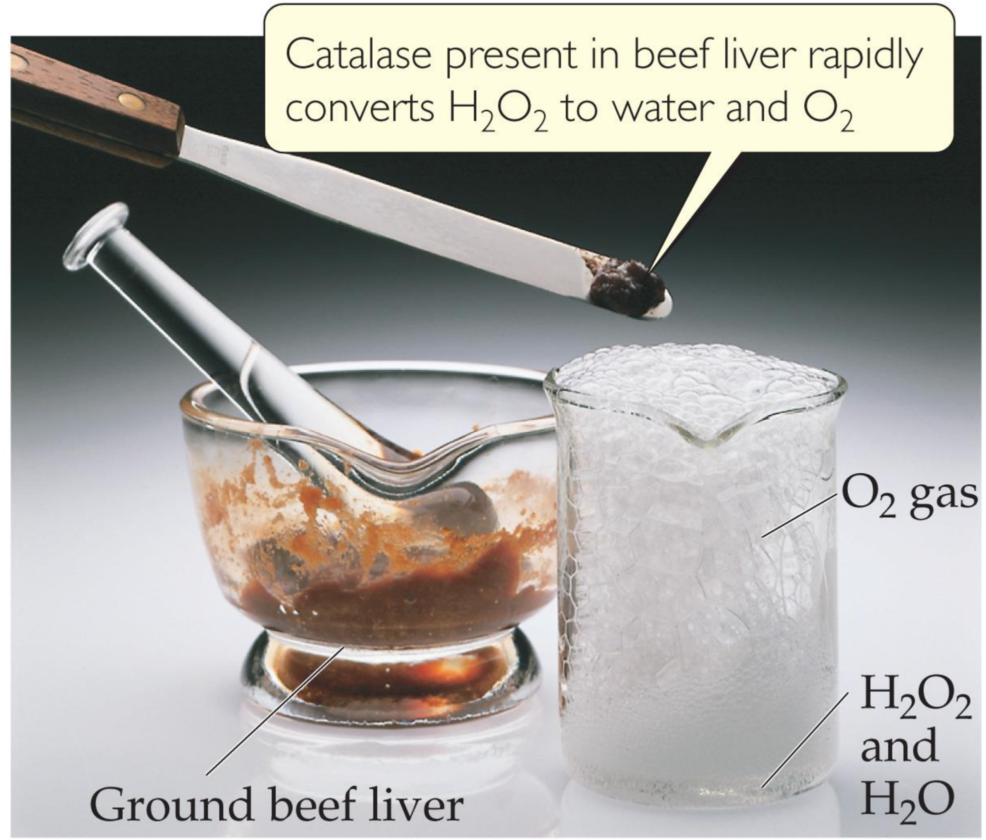
- The catalyst is in a different phase than the reactants.
- Often, gases are passed over a solid catalyst.
- The adsorption of the reactants is often the rate-determining step.



Enzymes

- **Enzymes** are biological catalysts.

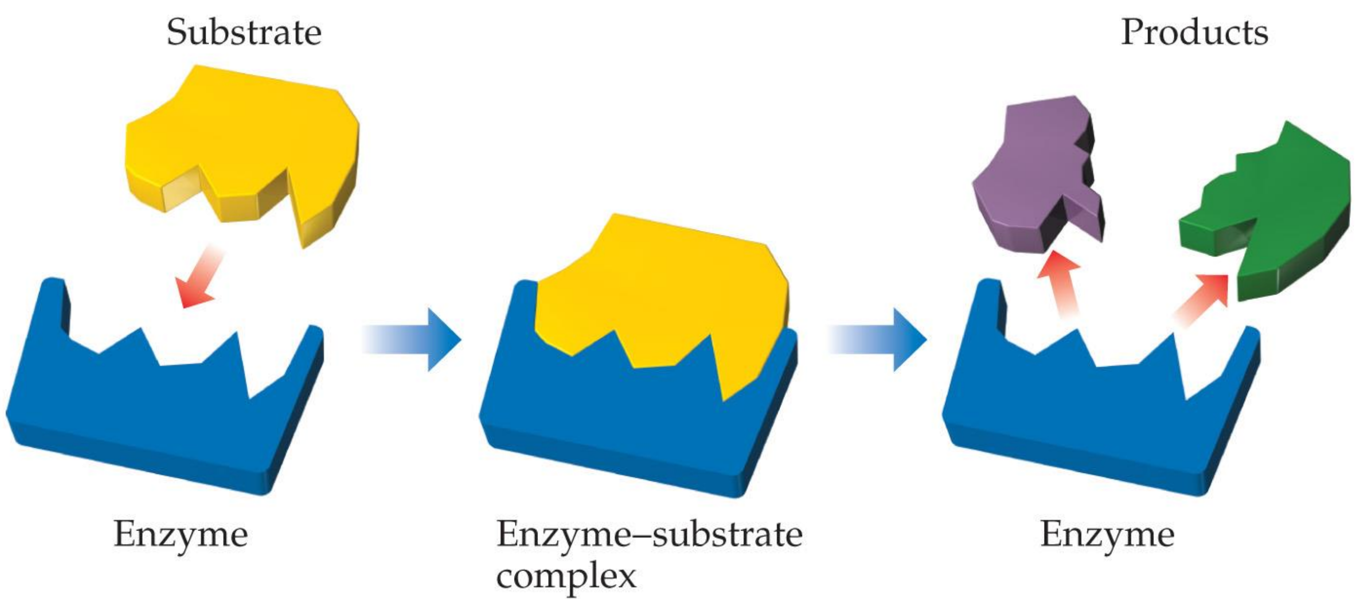
- They have a region where the reactants attach. That region is called the **active site**. The reactants are referred to as **substrates**.



Lock-and-Key Model

- In the enzyme–substrate model, the substrate fits into the active site of an enzyme, much like a key fits into a lock.

- They are specific.



Chemistry and Life

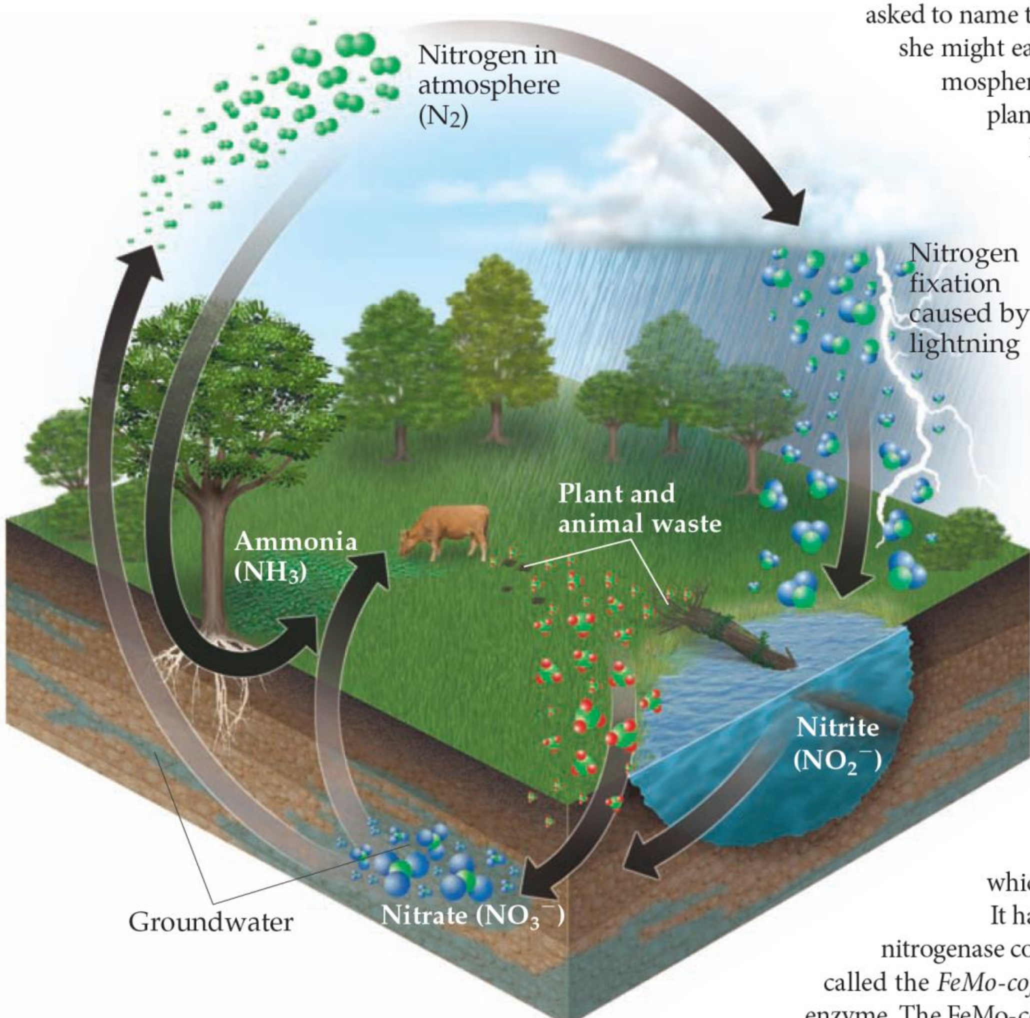
Nitrogen Fixation and Nitrogenase

Nitrogen is one of the most essential elements in living organisms, found in many compounds vital to life, including proteins, nucleic acids, vitamins, and hormones. Nitrogen is continually cycling through the biosphere in various forms, as shown in ◀ **Figure 14.29**. For example, certain microorganisms convert the nitrogen in animal waste and dead plants and animals into $N_2(g)$, which then returns to the atmosphere. For the food chain to be sustained, there must be a means of converting atmospheric $N_2(g)$ into a form plants can use. For this reason, if a chemist were asked to name the most important chemical reaction in the world, she might easily say *nitrogen fixation*, the process by which atmospheric $N_2(g)$ is converted into compounds suitable for plant use. Some fixed nitrogen results from the action of lightning on the atmosphere, and some is produced industrially using a process we will discuss in Chapter 15. About 60% of fixed nitrogen, however, is a consequence of the action of the remarkable and complex enzyme *nitrogenase*. This enzyme is *not* present in humans or other animals; rather, it is found in bacteria that live in the root nodules of certain plants, such as the legumes clover and alfalfa.

Nitrogenase converts N_2 into NH_3, a process that, in the absence of a catalyst, has a very large activation energy. This process is a *reduction* reaction in which the oxidation state of N is reduced from 0 in N_2 to -3 in NH_3. The mechanism by which nitrogenase reduces N_2 is not fully understood. Like many other enzymes, including catalase, the active site of nitrogenase contains transition-metal atoms; such enzymes are called *metalloenzymes*. Because transition metals can readily change oxidation state, metalloenzymes are especially useful for effecting transformations in which substrates are either oxidized or reduced.

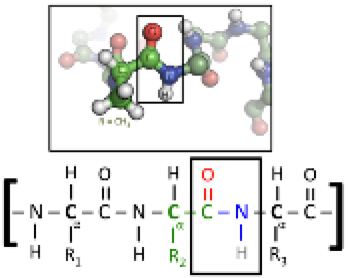
It has been known for nearly 40 years that a portion of nitrogenase contains iron and molybdenum atoms. This portion, called the *FeMo-cofactor*, is thought to serve as the active site of the enzyme. The FeMo-cofactor of nitrogenase is a cluster of seven Fe atoms and one Mo atom, all linked by sulfur atoms (▼ **Figure 14.30**).

It is one of the wonders of life that sim-



▲ **Figure 14.29** **Simplified picture of the nitrogen cycle.**

N_2



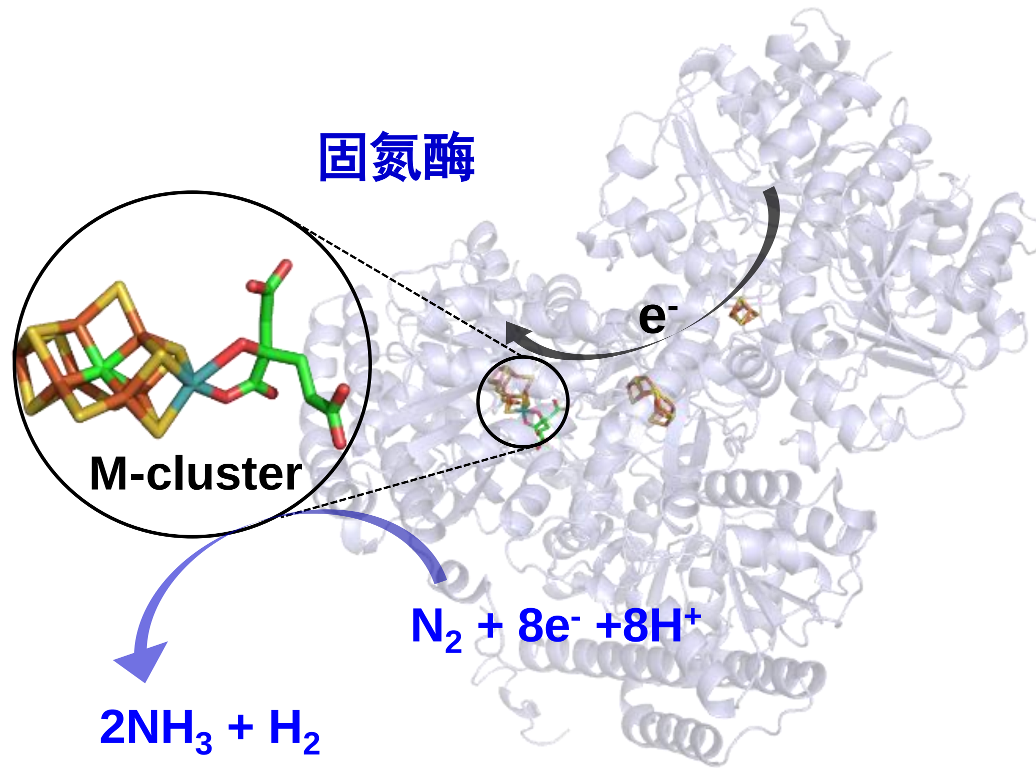
蛋白质

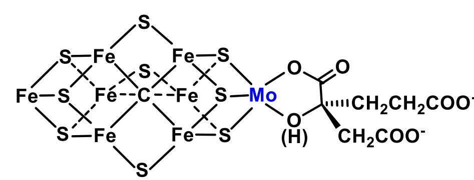
空气中含有
78%氮气N_2

:N⋮⋮N:

:N≡N:

根瘤菌



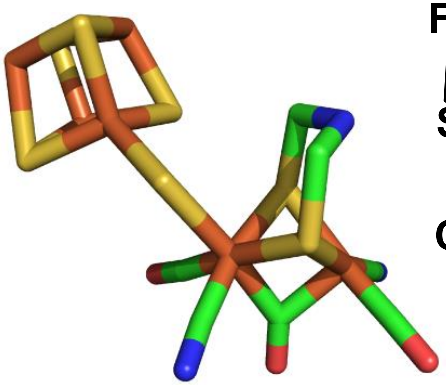


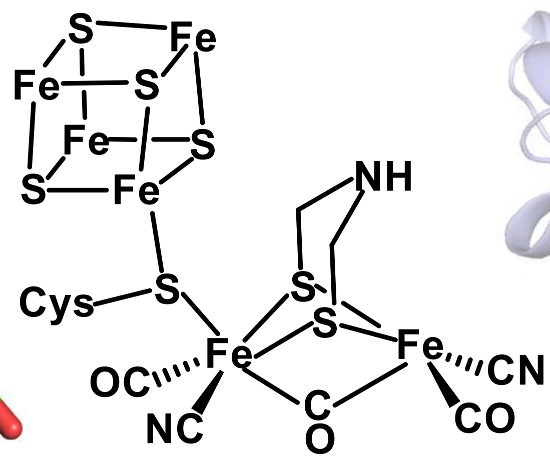
Fe_7MoS_9C

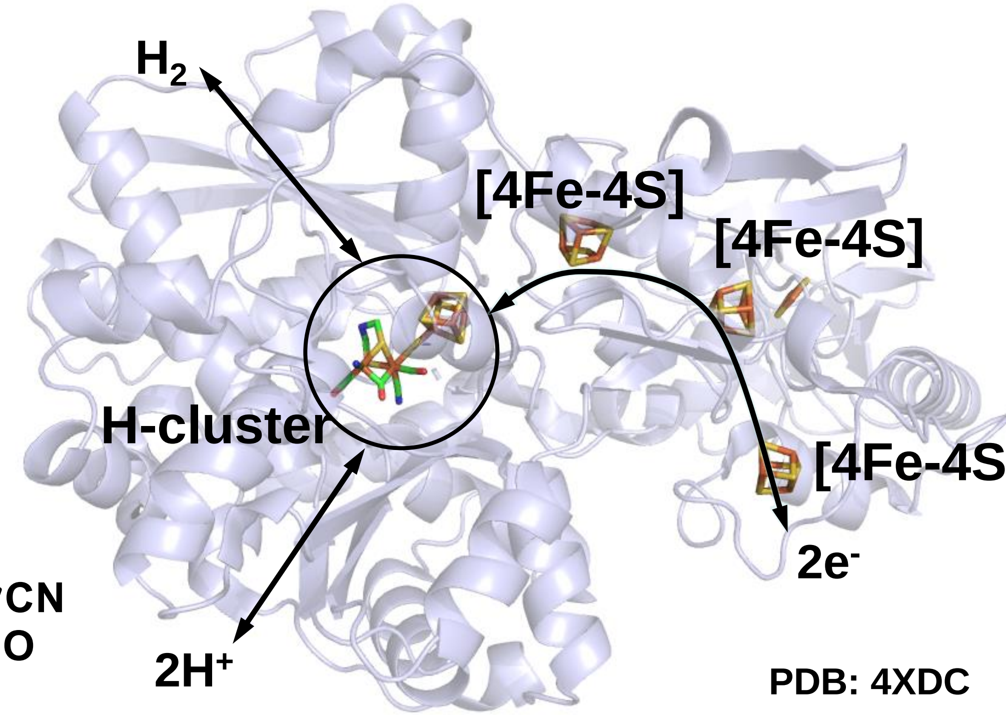
Chemical
Kinetics

[FeFe] hydrogenase

$$H_2 \rightleftharpoons 2H^+ + 2e^-$$







PDB: 4XDC

from Clostridium pasterianum, CpI

- **Reversibly convert hydrogen into protons and electrons;**
- **Extremely active in H_2 generation, with the TOF as high as 10000 s^{-1} at room temperature;**
- **Potential electrocatalysts in biofuel cells.**

Chemical

Seefeldt, et al., *Science*, 1998, 282, 1853.
Lubitz et al., *Chem. Rev.*, 2014, 114, 4081.

Key Equations

$\text{Rate} = -\frac{1}{a}\frac{\Delta[\text{A}]}{\Delta t} = -\frac{1}{b}\frac{\Delta[\text{B}]}{\Delta t} = \frac{1}{c}\frac{\Delta[\text{C}]}{\Delta t} = \frac{1}{d}\frac{\Delta[\text{D}]}{\Delta t}$	[14.4]	Definition of reaction rate in terms of the components of the balanced chemical equation $a\,\text{A} + b\,\text{B} \longrightarrow c\,\text{C} + d\,\text{D}$
$\text{Rate} = k[\text{A}]^m[\text{B}]^n$	[14.7]	General form of a rate law for the reaction $\text{A} + \text{B} \longrightarrow$ products
$\ln[\text{A}]_t - \ln[\text{A}]_0 = -kt$ or $\ln\frac{[\text{A}]_t}{[\text{A}]_0} = -kt$	[14.12]	The integrated form of a first-order rate law for the reaction $\text{A} \longrightarrow$ products
$\frac{1}{[\text{A}]_t} = kt + \frac{1}{[\text{A}]_0}$	[14.14]	The integrated form of the second-order rate law for the reaction $\text{A} \longrightarrow$ products
$t_{1/2} = \frac{0.693}{k}$	[14.15]	Relating the half-life and rate constant for a first-order reaction
$k = Ae^{-E_a/RT}$	[14.19]	The Arrhenius equation, which expresses how the rate constant depends on temperature
$\ln k = -\frac{E_a}{RT} + \ln A$	[14.20]	Logarithmic form of the Arrhenius equation